

Muhammad Sohel Rana
Department of Mathematics
University of North Texas at Dallas
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EDUCATION

Ph.D. in Mathematics (Data Science Track), University of Texas at Arlington – 2026

- Advisor: Dr. Keaton Hamm
- Dissertation: *Neighborhood Embeddings and Scalable Learning for Optimal Transport and Unbalanced Optimal Transport*

M.S. in Mathematics, Western Kentucky University – 2017

- Supervisor: Dr. Mark Robinson

M.S. in Applied Mathematics, University of Dhaka – 2008

B.S. in Mathematics, University of Dhaka – 2007

- Project Supervisor: Dr. Mohammad Ferdows

EMPLOYMENT

Assistant Professor, University of North Texas at Dallas	2026–Present
Graduate Research Assistant, University of Texas at Arlington	2023–2026
Graduate Teaching Assistant, University of Texas at Arlington	2022–2023
Lecturer, North South University	2022
Graduate Teaching Assistant, University of Texas at Arlington	2019–2021
Graduate Teaching Assistant, Western Kentucky University	2015–2017
Senior Officer, Pubali Bank Limited	2012–2015
Mathematics Teacher, Bangladesh International School and College	2012
Mathematics Teacher, Bangladesh International Tutorial	2011

RESEARCH INTERESTS

Optimal Transport; Unbalanced Optimal Transport; Machine Learning; Geometric Data Analysis; High-Dimensional Data Analysis; Dimensionality Reduction; Matrix Completion; Scientific Machine Learning.

PUBLICATIONS

Submitted Preprint

- ◊ Muhammad Rana, and Keaton Hamm, **Neighbor Embeddings Using Unbalanced Optimal Transport Metrics**. (Revising to resubmit). PDF
- ◊ Muhammad Rana, Phuong Trinh, Ryan Bui, and Keaton Hamm, On Wasserstein distance neighbor embeddings. (Working)

Peer reviewed papers

- ◊ Muhammad Rana, Abiy Tasissa, HanQin Cai, Yakov Gavriylov, and Keaton Hamm, **Recovering Wasserstein Distance Matrices from Few Measurements**. In Proceedings of the 2026 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), Barcelona, Spain, May 3–8, 2026. PDF

- ◇ Md. Mehedi Hasan, Mohammad Abdul Halim, Muhammad Sohel Rana, Salma Yeasmin Jannaty, and M. J. Uddin. **Nanofluid natural convection in trapezoidal annuli: Comparative effects of triangular, circular, and square inner walls.** *International Journal of Thermofluids (Elsevier)*, 2025. PDF
- ◇ Zahidur Talukder, Muhammad Rana, Keaton Hamm, and Mohammad Islam, **Empowering Clients: Self-Adaptive Federated Learning for Data Quality Challenges.** In Proceedings of the 2025 IEEE International Conference on Edge Computing and Communications (EDGE), Helsinki, Finland, July 7–12, 2025. PDF

Dissertation

- ◇ Muhammad Rana, **Neighborhood Embeddings and Scalable Learning for Optimal Transport and Unbalanced Optimal Transport.** (PhD Dissertation) PDF

AWARDS

- ◇ Mathematics Academic Excellence Scholarship, The University of Texas at Arlington
 - ⊙ Awarded annually to graduate students who have demonstrated outstanding academic excellence (approximately 70 graduate students in the department).
- ◇ Dr. R. Glenn and Virginia Powers Scholarship, Western Kentucky University
 - ⊙ Awarded to the top returning graduate student in the Department of Mathematics (approximately 20 graduate students).
- ◇ Honors Merit Scholarship, University of Dhaka
 - ⊙ Awarded to the top 2–3 students in the Department of Mathematics residing in Fazlul Huq Muslim Hall, based on undergraduate honors results.

PRESENTATIONS

Invited Talks

- ◇ *Empowering Clients: Self-Adaptive Federated Learning for Data Quality Challenges.* Teatime Event, Division of Data Science, The University of Texas at Arlington, Arlington, TX, February 21, 2025.
- ◇ *Optimal Transport-Based Dimensionality Reduction.* CodEx Student/Postdoctoral Series Mini-Conference (Virtual), May 27, 2025.

Poster Presentations

- ◇ *Recovering Wasserstein Distance Matrices from Few Measurements.* Discover 2026 Student Research Symposium, The University of Texas at Arlington, Arlington, TX, April 2026.

TEACHING EXPERIENCE

University of North Texas at Dallas
2026–Present

Assistant Professor

- MATH 3315: Linear Algebra and Vector Space Theory – Fall 2026
- MATH 1332: A Survey of Mathematics with Applications – Fall 2026
- MATH 1303: Fundamentals of Mathematics with Applications – Fall 2026

University of Texas at Arlington
2019–2023

Graduate Teaching Assistant

- MATH 1426: Calculus I – Fall 2019
- MATH 2425: Calculus II – Spring 2020, Fall 2020, Spring 2021, Fall 2022, Spring 2023
- STAT 1308: Elementary Statistical Analysis – Summer 2020

North South University
2022

Lecturer

- MATH 116: Precalculus (5 sections) – Spring 2022, Summer 2022

- MATH 120: Calculus and Analytic Geometry I (4 sections) – Spring 2022, Summer 2022

Western Kentucky University
2015–2017

Graduate Teaching Assistant

- MATH 136: Calculus I – Fall 2015
- MATH 137: Calculus II – Spring 2016, Fall 2016, Spring 2017, Fall 2017

PROFESSIONAL EXPERIENCE

Pubali Bank Limited, Bangladesh
2012–2015

Senior Officer

- Managed client accounts and supported retail banking operations.
- Processed and monitored loan applications, disbursements, and recoveries.
- Provided customer service while ensuring compliance with banking policies and procedures.

ACADEMIC SERVICES

Volunteering Experience

- UTA Calculus Bowl : Assisted with a playoff-style quiz competition focused on precalculus and calculus concepts for student teams – 2020, 2021, 2023, 2024

Peer Review Activities

- Peer Reviewer, American Journal of Undergraduate Research (AJUR), 2026.

TECHNICAL SKILLS

- **Programming:** Python, C, Fortran, Mathematica, $\LaTeX$
- **Python Libraries:** NumPy, Pandas, SciPy, scikit-learn, PyTorch, TensorFlow, Keras, Matplotlib, NLTK
- **Database:** PostgreSQL
- **Computational Methods:** Machine Learning, Scientific Computing, Numerical Analysis, Data Analysis